Which six customers, and what are the odds?

A pre-registered hybrid quantum–classical pipeline for the bank term-deposit campaign, reported with its negative results intact.

0.567655

P(≥3 OF 6 SUBSCRIBE)

39.7849

EXPECTED NET VALUE

C0081 C0031 C0079 C0170 C0327 C0182

SELECTED · COST 12/25 · CAPACITY 13/22

144 tests | seed 42 | 5 stages | 0 quantum wins claimed

Stages 0–6 · reproducible end to end

Five stages, one frozen protocol

Every learned statistic is fitted inside a fold. The selection rule was hashed before any model was trained.

STAGE 0–1

Classical baseline. 5×10 repeated stratified CV, 3 registered candidates, nested sigmoid calibration.

`ROC-AUC 0.6463`

STAGE 2

Quantum classifiers. ZZ fidelity kernel, data-reuploading VQC, plus the ablations that make the comparison fair.

`gate: rejected`

STAGE 3–4

Selection. One probability vector, top 10, 20-qubit QUBO, brute force, standard + Dicke/XY QAOA.

`optimum 39.7849`

STAGE 5–6

Outcome. 6-qubit circuit, exact Poisson-binomial, shot/noise/mitigation runs, traceability bundle.

`P(≥3) 0.567655`

The protocol is the deliverable. Test rows are never used for features, tuning, calibration, thresholds or model selection; the admission rule and its fingerprint were written into `PREREGISTRATION.md` before Stage 1 ran, and Stage 3 refuses to continue if the artifact it reads was produced under a different fingerprint.

Stage 1 — the classical bar to beat

CANDIDATE	ROC-AUC	PR-AUC	BRIER
GB tuned · n=100, d=2, η=0.03	0.6463 ± 0.0781	0.2984	0.0997
random forest	0.6301	0.2761	0.1050
gradient boosting (base)	0.6210	0.2687	0.1038
logistic regression L2	0.5761	0.2384	0.1085

50 folds · 60-row validation slices · 35 positives in 300 rows.

CALIBRATION DECISION

Nested sigmoid calibration made Brier **worse**: 0.0997 → 0.1010.

The pre-registered fallback applies, so Stage 3 ships **raw** probabilities. The bit is recorded before any test row is scored.

WHY THIS NUMBER MATTERS

0.6463 is the bar every quantum model is measured against — and the reason the pipeline reports a rejection instead of a story.

Stage 2 — the quantum models lost, and here is where

MODEL (4-D IN-FOLD PCA)	ROC-AUC	Δ VS GB	FOLDS WON
GB on the same 4-D features	0.5431	−0.103	—
VQC · 16 params · COBYLA	0.5691	−0.0772	13 / 50
ZZ fidelity kernel + SVC	0.5130	−0.1332	9 / 50
unentangled kernel (ablation)	0.4794	−0.167	6 / 50
classical RBF SVM, same features	0.4542	−0.192	4 / 50

THE DECISIVE ABLATION

GB drops **0.6463** → **0.5431** when restricted to the same 4-D projection.

78 %

of the ZZ kernel's 0.133 gap is feature compression, not the quantum map.

Entanglement is worth +0.0336 (28/50 folds) — suggestive, not significant. Both quantum models fail the gate with 0/10 repeats won, so no quantum score enters Stage 3.

Stage 3 — one vector, and a tie that decided slot 10

#	ID	P-HAT	BOOTSTRAP	TOP-10
1	C0081	0.718054	0.498 ± 0.242	84 %
2	C0031	0.703851	0.477 ± 0.251	81 %
3	C0079	0.654671	0.501 ± 0.256	81 %
4	C0170	0.208464	0.187 ± 0.092	36 %
5	C0327	0.203974	0.215 ± 0.112	48 %
6	C0183	0.200777	0.235 ± 0.116	57 %
7	C0182	0.186305	0.153 ± 0.083	20 %
8	C0193	0.178435	0.145 ± 0.093	14 %
9	C0349	0.169156	0.180 ± 0.111	29 %
10	C0359	0.169156	0.170 ± 0.101	27 %

TIE-BREAK IS LOAD-BEARING

C0349 and C0359 receive an **identical** score to double precision. The pre-registered rule (ascending `customer_id`) decides slot 9, not the model.

THE POINT ESTIMATE IS NOT THE ANSWER

0.718 → 0.498 under bootstrap. With 35 positives the model knows *who leads*, not *how likely*.

Stage 4 — an exact answer, and constraints that never bite

QUBO

$$\min -\Sigma\, w_i x_i + 40(\Sigma x - 6)^2 + 12(\Sigma c \cdot x + s_b - 25)^2 + 12(\Sigma h \cdot x + s_c - 22)^2$$

20 qubits · 10 decision + two 5-bit slack registers · 165 couplings

QUANTITY	BOUND	WORST CASE OVER ALL 210 SUBSETS
cardinality	= 6	6
cost	≤ 25	15
capacity	≤ 22	18

FINDINGS

- All **210** subsets are feasible, so the QUBO's feasible set is the whole weight-6 shell and the optimum is simply the **six largest w_i**.
- The budget and capacity rows **cannot** change the answer for this instance.
- QUBO ground state (−39.7849) **equals** the brute-force optimum; the run aborts if it does not.

Stated plainly: on this instance no optimiser — quantum or classical — can beat sorting six numbers. The pipeline says so instead of dressing it up.

Stage 4 — QAOA: encoding beats penalty strength

SOLVER	P	QUBITS	FEASIBILITY	P(OPT)	A
standard soft penalty	1	20	21.7 %	1.4e-4	0.549
standard soft penalty	3	20	19.9 %	2.2e-4	0.542
standard soft penalty	5	20	20.0 %	9.8e-4	0.546
Dicke + XY ring	1	10	100 %	0.0044	0.729
Dicke + XY ring	3	10	100 %	0.0175	0.813
Dicke + XY ring	4	10	100 %	0.0759	0.832
Dicke + XY ring	5	10	100 %	0.101	0.817

α = E[net value] / optimal net value over the exact output distribution.

WHAT THE NUMBERS SAY

- Soft penalties spend amplitude on low-cardinality states; more depth does not fix it (energy 2008 → 2554 vs ground −39.8).
- Dicke/XY holds Σx=6 exactly, and by p=4 its single most probable bitstring **is** the optimum: 838/8 192 shots at p=5.

Simulators verified against Qiskit/Aer on the full problems: max |ΔP| = 2.6e-16 (standard), 4.5e-16 (Dicke).

Stage 5 — six qubits, and every row is a real simulation

METHOD	SHOTS	P-HAT(≥3)	ERROR
exact Poisson-binomial	∞	0.567655	0
ideal statevector	∞	0.567655	2e-16
Aer sampling	8 000	0.580875	1.3e-2
Aer sampling	100 000	0.567060	5.9e-4
noisy Aer · 20 seeds	8 000	0.5570 ± 0.0049	−0.0106
readout-mitigated · 20 seeds	8 000	0.5664 ± 0.0059	−0.0012

The 8 000-shot draw really is 2.4σ high. It is reported as measured, not re-seeded until it looks tidy.

CIRCUIT

6 qubits · 6 Ry(2 arcsin √p) · **0 two-qubit gates** · 0 SWAPs

logical depth 1 → depth 4 / 24 one-qubit gates in {rz,sx,x,cx}

MITIGATION IS MEASURED, NOT ASSERTED

A 64×64 assignment matrix built from simulated calibration circuits + NNLS removes **89 %** of the readout bias.

Where the uncertainty actually lives

SOURCE	MEAN	SD	95 % INTERVAL
shot noise @ 8 192	—	0.0055	—
frozen selection (1000 draws)	0.3486	0.2180	[0.027, 0.748]
re-solved selection (1000 draws)	0.5046	0.2025	[0.147, 0.853]

Model uncertainty is **~40×** quantum shot noise. The frozen-selection mean sits below the point estimate because refitting shrinks extreme probabilities.

SELECTION STABILITY

Under bootstrap, the top-10 candidate set is identical in **2.5 %** of re-solves, and the six-customer optimum repeats in **0.5 %**.

Mean regret of the frozen pick: 10.1 of 39.78 (p95 32.8).

This is the project's main negative result: the decision is dominated by 300 rows of training variance, not by any quantum effect.

What we claim, and what we do not

CLAIMED

- A reproducible pipeline: 144 tests, seeded, one-command reproduce.
- An exact, verified answer: 6 customers, 39.7849 net value, 0.567655 success probability.
- Quantified decomposition of classical vs quantum contributions — including where quantum lost.

NOT CLAIMED

- No quantum advantage. The Stage 4 instance is solved exactly by 210-way enumeration and by sorting.
- No hardware results: simulators only (qiskit-aer 0.18.2).
- No test-set accuracy: the release contains no test labels.

DELIVERABLE

`submission.zip · results/traceability.md · deck/slides.pdf · reproduce.sh · 144 tests passing`